

LAPORAN PRAKTIKUM

UAS INFORMATION RETRIEVAL

**Eksplorasi Data Web Scraping Berbasis Scrapy dan Sistem
Informasi Pencarian Buku dengan Streamlit**

DOSEN PENGAMPU:

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PROGRAM STUDI (S1) SISTEM INFORMASI

FAKULTAS TEKNIK

UNIVERSITAS ABULYATAMA

ACEH BESAR

2026

BAB I: PENDAHULUAN

Perkembangan teknologi informasi menuntut ketersediaan data yang cepat dan terstruktur. Sering kali, informasi yang dibutuhkan publik tersebar di berbagai situs web dan belum tersaji dalam format basis data siap pakai. Untuk mengatasi tantangan ini, metode otomatisasi seperti Web Scraping menjadi solusi krusial dalam Information Retrieval.

Praktikum ini merancang sebuah sistem end-to-end yang dimulai dari proses ekstraksi data mentah dari internet menggunakan framework Scrapy, penyimpanan data terstruktur (JSON), hingga penyajian antarmuka berbasis web interaktif menggunakan framework Streamlit agar pengguna dapat mencari informasi buku secara efisien.

BAB II: TUJUAN KEGIATAN

Adapun sasaran utama dari pelaksanaan tugas ini meliputi:

1. Memahami mekanisme kerja framework Scrapy dalam melakukan crawling dan parsing halaman web secara otomatis.
2. Mampu mengubah data tidak terstruktur dari web target menjadi format data terstruktur berbentuk file JSON.
3. Mengembangkan aplikasi antarmuka web interaktif berbasis Python menggunakan Streamlit untuk mempermudah pencarian informasi.
4. Melakukan manajemen versi kode sumber menggunakan Git serta mempublikasikan aplikasi ke ranah publik secara online.

BAB III: TAHAPAN PELAKSANAAN DAN DOKUMENTASI

3.1. Inisialisasi Struktur Direktori Proyek

Deskripsi Kerja: Menyiapkan folder kerja lokal (misalnya folder Tugas IR) yang di dalamnya memuat direktori proyek Scrapy dan folder penyimpanan dataset.

```
book_crawler_app/
├── app.py                ← Aplikasi Streamlit (UI Search Engine)
├── data/
│   └── books.json        ← Data hasil crawling (1000 items)
├── scrapy_project/      ← Folder proyek Scrapy
│   ├── scrapy.cfg       ← File konfigurasi utama Scrapy
│   └── scrapy_project/  ← Paket/Modul Scrapy
│       ├── spiders/
│       │   └── books_spider.py ← Script spider crawler
│       ├── __init__.py
│       ├── items.py
│       ├── middlewares.py
│       ├── pipelines.py
│       └── settings.py
├── requirements.txt     ← Pustaka Python (streamlit & scrapy)
└── README.md            ← Dokumentasi, Identitas, & Link Akses
```

3.2. Pembuatan Kerangka Proyek Scrapy

Deskripsi Kerja: Menjalankan perintah terminal 'scrapy startproject scrapy_project' untuk membentuk kerangka modul dan file konfigurasi awal Scrapy.

```
Command Prompt
Microsoft Windows [Version 10.0.22621.755]
(c) Microsoft Corporation. All rights reserved.

C:\Users\HP>scrapy startproject scrapy_project
New Scrapy project 'scrapy_project', using template directory 'C:\Users\HP\AppData\Local\Programs\Python\Python310\Lib\site-packages\scrapy\template
s\project', created in:
  C:\Users\HP\scrapy_project

You can start your first spider with:
  cd scrapy_project
  scrapy genspider example example.com
```

3.3. Penyusunan Skrip Logika Ekstraksi (Spider)

Deskripsi Kerja: Menulis kode program di dalam file books_spider.py untuk mendefinisikan aturan parsing elemen target (judul, harga, rating, ketersediaan).

```
books_spider.py
1 import scrapy
2
3 class BooksSpider(scrapy.Spider):
4     name = 'books'
5     start_urls = ['http://books.toscrape.com/catalogue/page-1.html']
6
7     def parse(self, response):
8         for book in response.css('article.product_pod'):
9             yield {
10                 'title': book.css('h3 a::attr(title)').get(),
11                 'price': book.css('p::text').get(),
12                 # Mengambil status ketersediaan dengan rapi (misal: "In stock")
13                 'availability': book.css('p::text').re_first('In stock'),
14                 'rating': book.css('p::text').re_first('Star-rating (.*?)'),
15                 'link': response.urljoin(book.css('h3 a::attr(href)').get())
16             }
17
18         next_page = response.css('li.next a::attr(href)').get()
19         if next_page:
20             yield response.follow(next_page, self.parse)
```

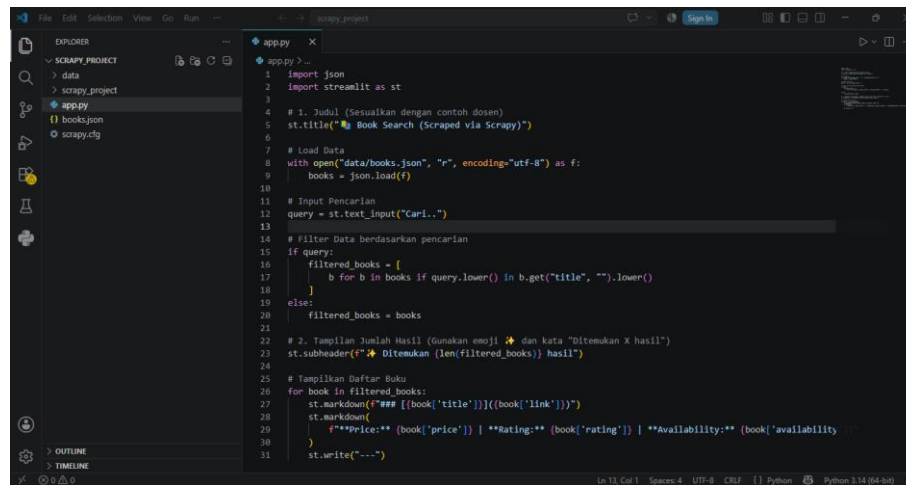
3.4. Proses Crawling & Penyimpanan Dataset JSON

Deskripsi Kerja: Mengeksekusi perintah crawl di terminal guna menyedot data dari web target dan menyimpannya secara otomatis ke format file books.json.

```
books.json
1 [
2   {
3     "title": "A Light in the Attic", "price": "€51.77", "availability": "In", "rating": "Three", "link": "htt
4     "title": "Tipping the Velvet", "price": "€53.74", "availability": "In", "rating": "One", "link": "https://
5     "title": "Sommers", "price": "€58.19", "availability": "In", "rating": "One", "link": "https://books.t
6     "title": "Sherry Objects", "price": "€42.82", "availability": "In", "rating": "Four", "link": "https://bc
7     "title": "Sapiens: A Brief History of Humankind", "price": "€54.23", "availability": "In", "rating": "Fiv
8     "title": "The Requiem Red", "price": "€22.65", "availability": "In", "rating": "One", "link": "https://bc
9     "title": "The Dirty Little Secrets of Getting Your Dream Job", "price": "€33.34", "availability": "In",
10    "title": "The Boys in the Boat: Nine Americans and Their Epic Quest for Gold at the 1936 Berlin Olympics", "p
11    "title": "The Black Maria", "price": "€52.15", "availability": "In", "rating": "One", "link": "https://bc
12    "title": "Starving Hearts (Triangular Trade Trilogy, #1)", "price": "€13.99", "availability": "In", "rati
13    "title": "Shakespeare's Sonnets", "price": "€20.66", "availability": "In", "rating": "Four", "link": "htt
14    "title": "Set Me Free", "price": "€17.40", "availability": "In", "rating": "Five", "link": "https://books
15    "title": "Scott Pilgrim's Previous Little Life (Scott Pilgrim #1)", "price": "€52.29", "availability": "I
16    "title": "Rip it Up and Start Again", "price": "€35.42", "availability": "In", "rating": "Five", "link":
17    "title": "Our Band Could Be Your Life: Scenes from the American Indie Underground, 1981-1991", "price":
18    "title": "Olio", "price": "€23.88", "availability": "In", "rating": "One", "link": "https://books.toscrap
19    "title": "Resurrection: The Best Science Fiction Stories 1880-1849", "price": "€37.59", "availability": "In
20    "title": "Libertarianism for Beginners", "price": "€51.37", "availability": "In", "rating": "Two", "link
21    "title": "It's Only the Himalayas", "price": "€45.17", "availability": "In", "rating": "Two", "link": "ht
22    "title": "In Her Wake", "price": "€12.84", "availability": "In", "rating": "One", "link": "https://books
23    "title": "How Music Works", "price": "€37.32", "availability": "In", "rating": "Two", "link": "https://bc
24    "title": "Foolproof Preserving: A Guide to Small Batch Jams, Jellies, Pickles, Condiments, and More: A Fo
25    "title": "Chase Me (Paris Nights #2)", "price": "€25.27", "availability": "In", "rating": "Five", "link:
26    "title": "Black Out", "price": "€34.53", "availability": "In", "rating": "Five", "link": "https://books
27    "title": "Birdsong: A Story in Pictures", "price": "€4.64", "availability": "In", "rating": "Three", "li
28    "title": "America's Cradle of Quarterbacks: Western Pennsylvania's Football Factory from Johnny Unitas to
29    "title": "Aladdin and His Wonderful Lamp", "price": "€33.13", "availability": "In", "rating": "Three", "l
30    "title": "Worlds Elsewhere: Journeys Around Shakespeare's Globe", "price": "€48.30", "availability": "In
31    "title": "Wall and Piece", "price": "€44.18", "availability": "In", "rating": "Four", "link": "https://bc
32    "title": "The Four Agreements: A Practical Guide to Personal Freedom", "price": "€17.66", "availability:"
```

3.5. Pembangunan Antarmuka Program Python (Streamlit)

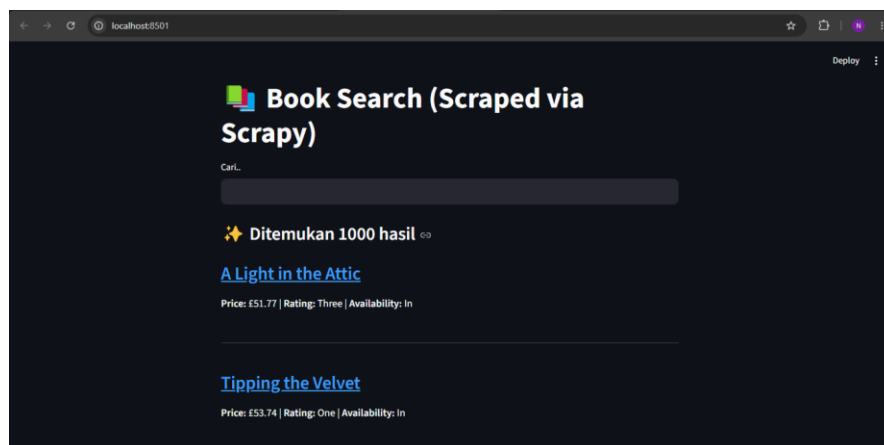
Deskripsi Kerja: Menyusun skrip app.py guna membaca file JSON dan merancang komponen UI pencarian interaktif.



```
1 import json
2 import streamlit as st
3
4 # 1. Judul (Sesuaikan dengan contoh dosen)
5 st.title("📖 Book Search (Scraped via Scrapy)")
6
7 # Load Data
8 with open("data/books.json", "r", encoding="utf-8") as f:
9     books = json.load(f)
10
11 # Input Pencarian
12 query = st.text_input("Cari...")
13
14 # Filter Data berdasarkan pencarian
15 if query:
16     filtered_books = [
17         b for b in books if query.lower() in b.get("title", "").lower()
18     ]
19 else:
20     filtered_books = books
21
22 # 2. Tampilkan Jumlah Hasil (Gunakan emoji 📖 dan kata "Ditemukan X hasil")
23 st.subheader(f"📖 Ditemukan {len(filtered_books)} hasil")
24
25 # Tampilkan Daftar Buku
26 for book in filtered_books:
27     st.markdown(f"### [{book['title']}] [{book['link']}]")
28     st.markdown(
29         f"💰 Price: {book['price']} | ⭐ Rating: {book['rating']} | 📦 Availability: {book['availability']}"
30     )
31     st.write("-----")
```

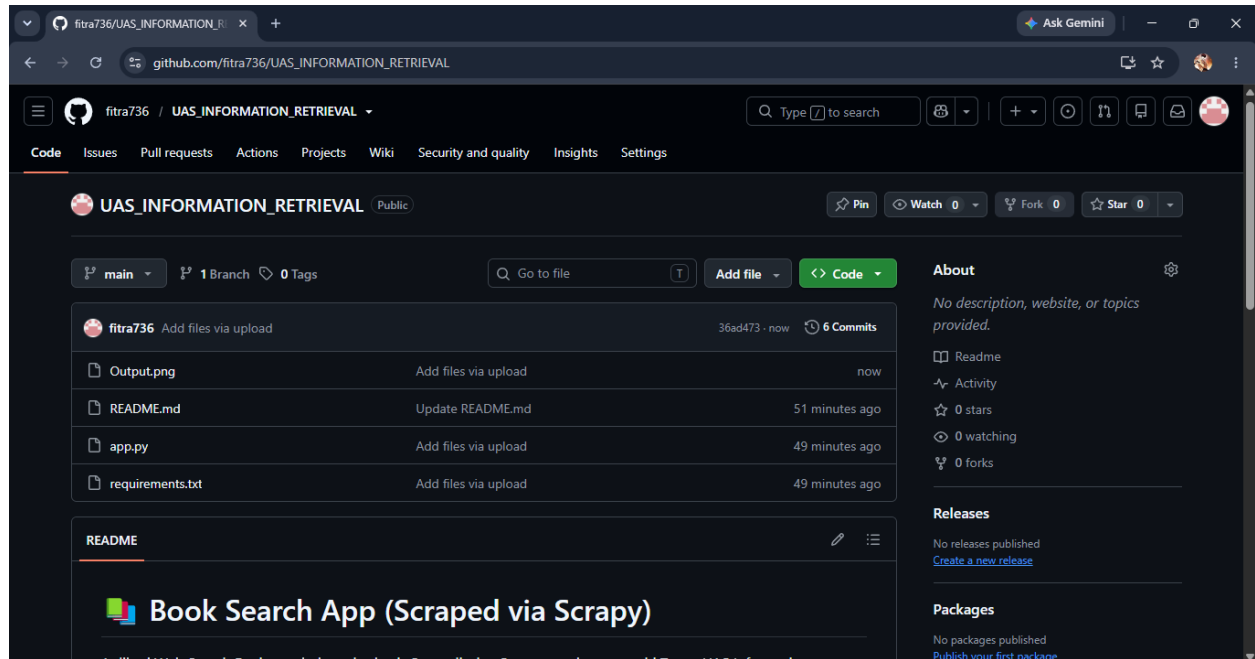
3.6. Validasi Aplikasi pada Server Lokal

Deskripsi Kerja: Menjalankan perintah 'streamlit run app.py' dan menguji sistem pencarian secara lokal melalui browser pada alamat localhost:8501.



3.7. Publikasi Repository & Deployment Web Online

Deskripsi Kerja: Mengunggah kode program ke repository GitHub publik serta menghubungkannya ke layanan deployment agar dapat diakses via internet.



BAB IV: PENUTUP DAN KESIMPULAN

Berdasarkan seluruh rangkaian proses implementasi yang telah dikerjakan, dapat ditarik kesimpulan bahwa:

- Kombinasi framework Scrapy dan Streamlit sangat handal dan efisien untuk membangun sistem temu kembali informasi (Information Retrieval) secara mandiri.
- Proses ekstraksi data web terbukti mempermudah pengumpulan data masif ke dalam format terstruktur yang siap diolah lebih lanjut.
- Seluruh tahapan dari pembangunan kode lokal hingga publikasi online telah berhasil divalidasi dan berjalan dengan baik.

Github :

https://github.com/fitra736/UAS_INFORMATION_RETRIEVAL